Principal Software Engineer - Radio Frequency

Clearance Level: Public Trust (Secret Eligible)

US Citizenship: Required

Job Classification: 1099/Consultant ($150 - $200 per hour)

Location: Remote

Years of Experience: 10+ years of relevant experience

Education Level: BS Degree — experience may be considered in place of education requirements.

Briefly Describe the Work:

GITI is seeking a Principal Software Engineer to support Cyber Operations Research and Development as the technical lead for production software development on a passive RF emitter identification and network analysis from real-time sensor data streams. The candidate will own the architecture, implementation, and delivery of the production pipeline — a stream ingestion, rollup, and post-processing system that operates on NDF (Network Description File) data produced by TDMA network sensors in dense, contested RF environments. The candidate will lead a small team of senior software engineers, coordinate closely with the program technical lead and AI/ML researchers to drive Cyber Operations software from prototype to production-quality, resource-efficient components deployable on tactical edge hardware. This is a hands-on technical leadership role: the Principal Engineer writes code, makes architecture decisions, and is accountable for pipeline performance and reliability in support of real world cyber operations.

Responsibilities:

Own the architecture and implementation of the production software pipeline, including stream ingestion, rollup, database write, and batch post-processing components
Lead a team of Senior Software Engineers in support of real world cyber operations; assign work, conduct code reviews, enforce quality standards, and provide technical mentorship
Establish and maintain disciplined software engineering practices: versioning, CI/CD pipelines, unit and integration testing, and documentation standards
Design and evaluate database and storage architecture for the tactical system and research enclave environments
Collaborate with the program technical lead to translate research findings and batch optimization algorithms into production pipeline components
Evaluate and benchmark Python pipeline performance on tactical-box-spec hardware; identify bottlenecks and lead porting of mature components to Rust or C for edge deployment

Manage and coordinate the tactical system VM environment and stream simulation infrastructure; ensure research VM is not disrupted by development activity
Define and enforce stream interface contracts between the ingestion layer, database, and downstream consumers
Evaluate emerging technologies (e.g., DuckDB/Parquet, Polars, message queues) against program requirements and recommend adoption decisions to the technical lead
Maintain the program's GitLab repository structure, branching strategy, and release management
Produce clear technical documentation including architecture decision records, interface specifications, and deployment guides
Support technical reviews and provide written inputs for sponsor deliverables as directed by the program technical lead
Expert-level career professional with broad and deep application of software engineering principles across the full development lifecycle. Exercises independent judgment in evaluating methods, techniques, and approaches; identifies and resolves complex technical problems with significant program impact. Provides technical leadership and direction to other engineers. Bachelor's (or equivalent) with 10+ years of experience.

Required Skills:

Demonstrated experience leading a software engineering team on a production data pipeline or streaming system; ability to set technical direction and mentor junior engineers
Expert-level Python development, including stream processing, multi-threaded/async architectures, and performance profiling
Proficiency in one or more compiled or systems languages (Rust, C, C++, or Go) for performance-critical components; experience porting Python to compiled targets
Hands-on experience designing and implementing relational database schemas and write-intensive data pipelines (MySQL, PostgreSQL, or equivalent)
Experience parsing binary serialization formats such as FlatBuffers or Protocol Buffers in a production context
Demonstrated ability to benchmark and optimize pipeline throughput on resource-constrained hardware or cloud environment
Strong proficiency with Linux system administration, remote server management via SSH, and air-gapped development environments
Experience architecting multi-consumer data systems where a single write path must serve concurrent display, analytics, and batch processing readers
Proficient in disciplined software engineering practices: GitLab/Git, CI/CD pipeline design, test-driven development, and code review
Excellent written and oral communication skills; ability to produce architecture decision records and technical documentation for both engineering and leadership audiences
Desired Skills:

Experience with TNS (Target Network System) sensor data formats and NDF ICD specifications

Familiarity with TDMA network protocols, time-division access architectures, and passive RF signal processing concepts
Experience deploying and operating software on tactical edge hardware co-located with a sensor system
Experience with lightweight stream or message queue architectures (ZeroMQ, RabbitMQ, or equivalent)
Experience with Polars or DuckDB for high-performance analytical workloads and write-once/read-many storage patterns
Experience with LLM-assisted software development tools (e.g., Claude Code, GitHub Copilot, JetBrains AI Assistant, or equivalent); demonstrated ability to use AI tools productively for code generation, refactoring, and test case development while maintaining engineering judgment and code quality standards
Familiarity with AI/ML model inference integration — ability to incorporate batch optimizer outputs into the production pipeline without requiring ML expertise
Experience with browser-based data visualization or reporting tools (React, D3, or equivalent) as a consumer of pipeline output
Experience with Jupyter Notebooks and research enclave environments; ability to bridge from research prototype to production code
Experience with FlatBuffers binary stream replay and simulation infrastructure for pipeline testing
Familiarity with Rust toolchain and ecosystem for systems-level development on Linux
Relevant Certifications:

Certifications in software engineering, computer science, or related fields (e.g., Certified Software Development Professional (CSDP); Certified Secure Software Lifecycle Professional (CSSLP); Red Hat Certified Engineer (RHCE); C++ Certified Professional Programmer (CPP); Professional Software Developer Certification (PSD); etc.)
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